AI Training Day 02

01

Hour 01 — Introduction to AI

02

Hour 02 — Basics of Machine Learning

03

Hour 03 — Key AI Concepts

04

Hour 04 — The AI Development Process

05

Hour 05 — Computer Vision Basics

06

Hour 06 — Basic English for AI



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HOUR 01

Introduction to Artificial Intelligence

What Is AI?

Artificial Intelligence is a branch of Computer Science concerned with creating machines that perform tasks requiring human-like intelligence. The term was coined by John McCarthy in 1956. AI systems analyze data, reason through problems, and make decisions — sometimes with no human involvement at all.

Four Main Approaches to AI

Think Like Humans

Cognitive modeling — replicating how the human mind works

Think Rationally

Logic-based reasoning and rules of inference

Act Like Humans

Turing Test — a machine indistinguishable from a human

Act Rationally

Rational agents that achieve the best possible outcome

Categories of AI by Capability

1

🤖 Super AI

Hypothetical — surpasses human intelligence in all aspects, including creativity and emotion

2

🧠 General AI (AGI)

Can perform any intellectual task a human can — not yet achieved

3

🎯 Narrow AI

Excels at one specific task — e.g., chess engines, image recognition, chatbots

What Can AI Do Today?

- Face recognition & computer vision
- Natural language processing & machine translation
- Medical diagnosis in specialized domains
- Autonomous vehicles & robotic planning
- Personalized recommendations & fraud detection

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HOUR 02

Basics of Machine Learning

Machine learning is the backbone of modern AI — it enables computers to learn from data and improve over time without being explicitly programmed for every scenario.

📘 Supervised Learning

The model learns from **labeled data** — every input has a known correct output. Like a teacher guiding a student with answer keys.

- Algorithm: trains on input-output pairs
- Tasks: Classification & Regression
- Examples: spam detection, house price prediction, image classification

🔍 Unsupervised Learning

The model works with **unlabeled data**, discovering hidden patterns and structures on its own — no predefined answers.

- Algorithm: finds structure without supervision
- Tasks: Clustering, Dimensionality Reduction, Association
- Examples: customer segmentation, product recommendations, anomaly detection

🎮 Reinforcement Learning

An agent learns by **trial and error**, receiving rewards for good actions and penalties for bad ones — improving through interaction.

- Algorithm: learns via a reward function
- Tasks: Decision-making & control problems
- Examples: game-playing AI, robotics, self-driving navigation

Core Building Blocks of Every ML System

→ Training Data

The examples the model learns from — quality and diversity are critical

→ Algorithm

The mathematical method that processes data and detects patterns

→ Model

The output of training — used to make predictions on new, unseen data

Key Terms to Know

Overfitting

Model memorizes training data but fails on new data — too specialized

Underfitting

Model is too simple to capture meaningful patterns — too general

Training vs. Testing

Data split between building the model and evaluating its real-world performance

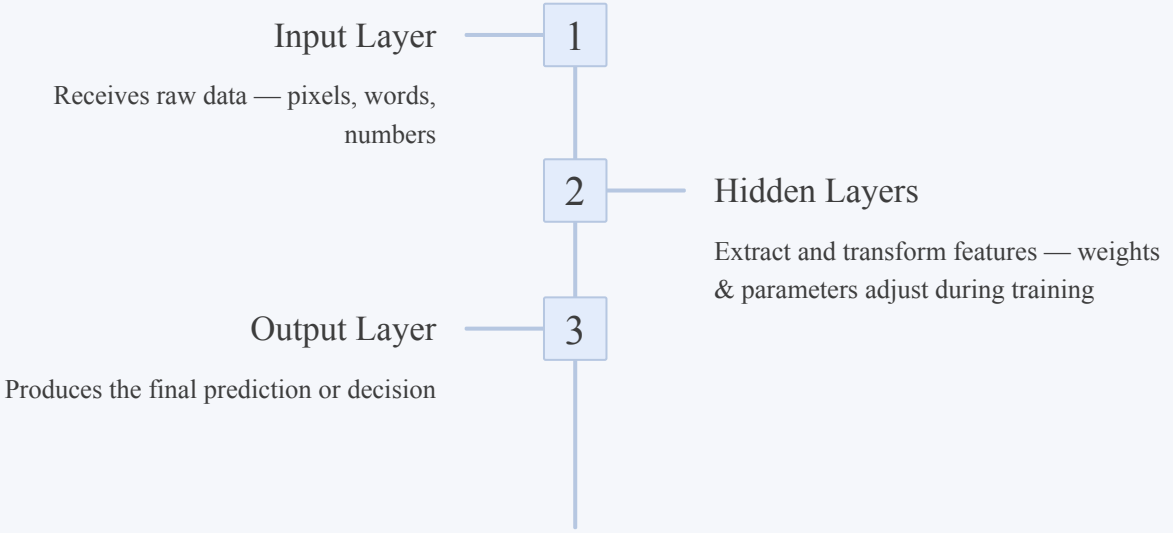
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HOUR 03

Key AI Concepts

Neural Networks & Deep Learning

Inspired by the human brain, neural networks are layers of interconnected nodes (neurons) that process data. **Deep Learning** refers to neural networks with many layers — the more layers, the more complex patterns they can learn.



Natural Language Processing (NLP)

NLP enables machines to understand and generate human language. Key tasks include text classification, sentiment analysis, machine translation, and question answering. Modern large language models (LLMs) like GPT are built on NLP foundations.

Essential AI Concepts at a Glance

Tokens & Parameters

Text is broken into tokens; model parameters control how inputs map to outputs — scale affects quality and cost

Hallucination

AI generates confident but incorrect or fabricated information — human verification is always essential

Prompt Engineering

Crafting effective instructions using role, task, context, constraints, tone, and output format to guide AI responses

Generative AI

Models that create new content — text, images, audio, video — rather than just classifying existing data

Bias & Fairness

AI systems can reflect biases in training data — ethical AI requires diverse data and continuous monitoring

AI Ethics & Governance

Data privacy, security, transparency, and accountability are critical responsibilities when deploying AI

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HOUR 04

The AI Development Process

Building an AI system is a structured, iterative lifecycle — not a single event. Each stage feeds the next, and teams revisit phases as new data and results emerge.

01

Define the Problem

Identify the business challenge, set clear objectives, and determine whether AI is the right solution. Vague requirements lead to costly rework in later stages.

02

Collect & Prepare Data

Gather raw data from sensors, databases, APIs, and external datasets. Clean, normalize, and label data — this is often the most time-consuming phase. Data quality directly determines model quality.

03

Select & Build the Model

Choose algorithms suited to the problem (e.g., regression, decision trees, neural networks). Design the architecture and configure hyperparameters for training.

04

Train & Evaluate

Feed prepared data into the model and adjust parameters until performance meets targets. Evaluate using metrics like accuracy, precision, recall, and AUC-ROC.

05

Deploy & Monitor

Integrate the model into production via APIs or cloud services. Continuously monitor for performance drift, update with new data, and maintain ethical compliance.

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HOUR 05

Computer Vision Basics

Computer vision is a field of AI that enables machines to interpret and understand the visual world — processing images and videos the same way humans use their eyes and brain.

Core Computer Vision Tasks

🖼️ Image Classification

Assigns a single label to an entire image — e.g., "this image contains a cat." Uses a classifier (deep neural network) to tag the whole image.

📦 Object Detection

Locates and classifies multiple objects within an image by drawing bounding boxes around each. Answers "what is it?" and "where is it?"

🧩 Image Segmentation

Identifies which exact pixels belong to each object — more precise than detection. Used in medical imaging and autonomous driving.

How It Works: The Pipeline

1. **Image Input**
   Camera / video / digital file
2. **Feature Extraction**
   Detect edges, shapes, patterns using CNN layers
3. **Prediction**
   Model outputs labels, bounding boxes, or pixel maps

Real-World Applications

Autonomous Vehicles

Detect pedestrians, road signs, lane markings, and obstacles in real time to navigate safely

Medical Imaging

Analyze X-rays, MRIs, and CT scans to detect tumors, fractures, and abnormalities

Security & Surveillance

Face recognition, crowd analysis, and threat detection in real-time video feeds

Manufacturing & QA

Automated visual inspection on production lines — catching defects faster than humans

Key Algorithms to Know

- **CNN (Convolutional Neural Network)** — the foundational architecture for image tasks
- **YOLO (You Only Look Once)** — fast, real-time multi-object detection in a single pass
- **Vision Transformers (ViT)** — applying transformer attention mechanisms to image patches

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HOUR 06

Basic English for AI

Understanding AI requires fluency in its vocabulary. Mastering these essential English terms will help you read research papers, documentation, and communicate clearly with international AI teams.

Foundational AI Vocabulary

Algorithm

A set of step-by-step mathematical instructions a computer follows to solve a problem or learn from data

Model

The trained output of a machine learning process — used to make predictions or decisions on new inputs

Dataset

A structured collection of data (text, images, numbers) used to train, validate, or test an AI model

Training

The process of feeding data into an algorithm so it learns patterns and adjusts its internal parameters

Inference

Using a trained model to make predictions on new, unseen data — the "live" usage phase of a model

Deployment

Releasing a trained AI model into a real production environment where it serves actual users or systems

Technical Terms in Context

Feature

An individual measurable property of the data — e.g., age, pixel brightness, word frequency

Label / Target

The correct answer or output value associated with a training data sample in supervised learning

Accuracy

The percentage of correct predictions made by a model out of all predictions — a basic performance metric

Epoch

One complete pass through the entire training dataset — models typically train for many epochs

Benchmark

A standard dataset or test used to compare and evaluate the performance of different AI models

API (Application Programming Interface)

A way for software applications to communicate — AI models are often accessed and deployed via APIs

Tips for Reading AI Documentation in English

- Look up every unfamiliar acronym immediately (NLP, CNN, LLM, RAG, etc.)
- Read paper abstracts first — they summarize the entire contribution
- Use context clues: formulas and diagrams often explain the surrounding English text
- Practice writing short prompts and instructions — precision in English improves AI output quality

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